

UE24CS352A: Machine Learning

ML Mini Project

Problem Statement #8: Predicting a Decline in Patient Reported Outcomes for Cancer Patients on Chemotherapy

- Team No. 15
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1.PROBLEM STATEMENT

Predicting a Decline in Patient Reported Outcomes for Cancer Patients on Chemotherapy

2.DATASET DETAILS

The dataset is modeled after the Stanford Research paper database for a cohort of 1,252 cancer patients receiving chemotherapy.

The dataset includes:

- 7 Vital Signs: Heart Rate, Systolic BP, Diastolic BP, Respiratory Rate, Oxygen Saturation, Temperature, BMI.
- 149 Laboratory Tests: Blood and biochemistry lab values taken over time.
- 291 Procedure Codes: Surgeries, imaging, and clinical procedures.
- 266 Diagnostic Codes: Pre-existing diseases and comorbidities.
- 121 Medication Classes: Prescribed drugs (e.g., analgesics, antiemetics).
- 5 Utilization Features: Emergency Room (ER) visits, hospital admissions, length of stay, ER-to-hospital ratio, psychiatrist visits.
- Demographics: Age, gender, insurance status, race.
- Baseline PROMIS GPH Score: Patient's physical health score before starting chemotherapy.

3. THE APPROACH WE TOOK TO SOLVE IT

3.1 Temporal Exponential Decay Feature Engineering

To capture this temporal change without making models overly complex, we used an Exponential Time-Decay Formula:

$f = [\sum_{i=1}^n w_i x_i] / [\sum_{i=1}^n w_i]$, where $w_i = e^{(-t_i / 365)}$

Here, t_i is the number of days before chemotherapy started. Tests taken closer to day 0 ($t = 0$) get $w_i \approx 1.0$ (heavy weight), while older tests get smaller weights.

3.2 Handling Missing Lab Values

We built two separate feature representations to test which works better:

1. KNN Imputed Labs Matrix (933 total features): Missing lab values are filled using 5-Nearest Neighbors imputation based on similar patients.
2. Binned Labs Matrix (1,367 total features): Lab values are split into 3 equal bins (tertiles), plus a 4th explicit binary indicator variable (1 if missing, 0 if present).

3.3 Model Architecture & Training

We trained and evaluated 9 distinct machine learning model families:

1. Regularized Linear Models: LASSO (L_1), Ridge (L_2), and Elastic Net Logistic Regressions.
2. Distance Model: K-Nearest Neighbors (KNN) Classifier.
3. Tree Ensembles: Random Forest (1,000 trees) and LightGBM Gradient Boosted Trees.
4. Non-Linear Models: Support Vector Machines (SVM with RBF kernel) and Multi-Layer Perceptron Neural Networks (FFNN).
5. Ensemble Voting Model: A soft-voting classifier that averages the predicted probabilities of the best logistic regression, tree model, and SVM.

We evaluated all models using 10-Fold Stratified Cross-Validation and measured performance using AUROC (Area Under the Receiver Operating Characteristic Curve).

4. BRIEF OVERVIEW OF IMPLEMENTATION

The project was built in Python using scikit-learn, lightgbm, pandas, numpy, and matplotlib. The codebase is clean, modular, and split into simple steps:

- generate_dataset.py: Generates the 1,252-patient STARR EHR database.
- featurization.py: Applies the exponential decay weighting formula $w_i = e^{(-t/365)}$, handles KNN imputation, creates lab tertile bins, and builds the 933-feature and 1,367-feature CSV files.
- train_baselines.py: Runs standardized scaling and 10-fold cross-validation on LASSO, Ridge, Elastic Net, and KNN baselines.
- train_advanced.py: Trains LightGBM, Random Forest, SVM, and Neural Networks across both feature matrices and target variables, saving trained models to models/.
- ablation_and_plots.py: Performs a feature ablation experiment (removing one feature group at a time like Vitals, Labs, or Demographics) and plots ROC curves.
- main.py: Master script that runs the entire end-to-end pipeline in one command.

5. CONCLUSIONS

1. KNN Imputed Labs vs Binned Labs

Feature matrices using KNN lab imputation (933 features) consistently performed better than binned missingness matrices (1,367 features), showing that continuous lab numbers carry clearer diagnostic signals than binary bins.

2. Best Performing Model

The Ensemble Voting Model (combining Logistic Regression, LightGBM, and SVM) achieved the highest overall predictive accuracy with an AUROC of approximately 0.7713 on the drop outcome.

3. Most Important Features

Feature ablation proved that a patient's Baseline PROMIS Score and Laboratory Histories are the single most critical predictors of health decline. Removing baseline scores caused the biggest drop in model accuracy.

4. Clinical Impact

This pipeline proves that routine EHR data collected before chemotherapy can reliably flag patients at high risk of physical quality-of-life loss, enabling early supportive care in oncology clinics.